

Course Code: CS1929		
Name of the Programme: BCA (Hons.)		
Name of the Course: Structured Innovation with Design Thinking		
Semester: I		
Course Credit	No. of Hours Per Week (L + T + P)	Total No. of Teaching Hours
2	1:0:2	45
Course Objectives:		
a) Foster an appreciation for finding value in everyday objects and surroundings		
b) Cultivate creativity and aesthetic sensibility		
c) Introduce the concept of products with purpose		
d) Guide students in generating and developing simple, meaningful product ideas		

Course Outline (Syllabus of the Course)

Module - 1 (Design thinking for creative problem-solving)	No. of Hours
	10 Hrs.
Introduction to Design Thinking: Overview of design thinking principles and stages. Empathy in Design: Techniques for understanding and empathizing with users, including user interviews and persona creation. Storytelling as a Tool: Using storytelling to convey user needs and problem statements. Case Studies: Analyzing successful design thinking projects that started with empathy. Workshops: Hands-on activities to practice empathic engagement with hypothetical user scenarios.	
Module - 2 (Concepts for sustainable prototype development)	No. of Hours
	10 Hrs.
Basics of Recycling and Upcycling: Definitions, differences, and environmental impact. Triple Constraints in Product Design: Exploring how time, scope, and budget affect product development. Prototype Development: Step-by-step guidance on creating prototypes using upcycled materials. Sustainability in Design: Discussing the importance of sustainable practices in modern design. Practical Exercises: Designing and building simple products using recycled materials within defined constraints.	
Module - 3 (Existing products focusing on market aesthetics)	No. of Hours
	10 Hrs.

Understanding Product Aesthetics: How aesthetics influence consumer behavior and product design. Detailed Product Analysis: Dismantling and evaluating existing products to understand their design and functionality. Enhancement Techniques: Methods for improving product design, focusing on aesthetics and functionality. Consumer Feedback: Gathering and utilizing consumer feedback to guide redesign efforts. Reconstruction Projects: Students reconstruct an existing product with improved aesthetics and functionality.	
Module - 4 (Innovation vs. Invention)	No. of Hours
	10 Hrs.
Exploring the difference and focusing on innovation within existing frameworks. Analyzing Market Trends: How to use market research to identify opportunities for innovation. Creative Adaptations: Techniques for creative thinking and adaptation of existing products. Prototyping with Purpose: Building prototypes that address specific user needs or market gaps. Case Studies and Discussions: Reviewing examples of successful innovations in existing product lines.	
Module - 5 (Communicate project processes and outcomes)	No. of Hours
	10 Hrs.
The Art of Storytelling: Techniques for crafting compelling narratives around product development. Presentation Skills: Tips and strategies for effective public speaking and presentation. Visual and Digital Tools: Using visual aids, digital presentations, and storytelling software to enhance communication. Engaging Stakeholders: How to pitch and explain projects to different audiences (investors, customers, peers). Project Documentation and Feedback: Methods for documenting the development process and incorporating feedback for improvement.	

Course Outcomes: After completing the course, the students will be able to: -
a) Apply design thinking for creative problem-solving
b) Synthesize concepts for sustainable prototype development
c) Enhance existing products focusing on market aesthetics
d) Innovate within frameworks to create new prototypes

Text Books

1	Knapp, J., Zeratsky, J., & Kowitz, B. (2016). <i>Sprint: How to solve big problems and test new ideas in just five days</i> (1st ed.). Simon & Schuster. ISBN 978-1501121746
2	Mootee, I. (2014). <i>Design thinking for strategic innovation: What they can't teach you at business or design school</i> (1st ed.). Wiley. ISBN 978-1118620120

Reference Books	
1	Liedtka, J., & Ogilvie, T. (2011). <i>Designing for growth: A design thinking toolkit for managers</i> (1st ed.). Columbia University Press. ISBN 978-0231158381
2	Soni, P. (2020). <i>Design your thinking</i> (1st ed.). Penguin Random House India Private Limited. ISBN 978-0670094097

Laboratory Component

30 Hrs

S.No	Activity	
1	Day 1: Paper Prototyping	Prototyping paper material & tools
2	Day 2: Upcycling	-
3	Day 3: Product study & variation	Products in market from various domains
4	Day 4: Game Design for a purpose	-